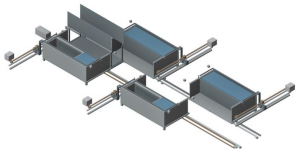
Seven feeder concepts · one comparison boundary

Preserved FDR09-R1 geometry · four mixed 1U/3U payload surrogates

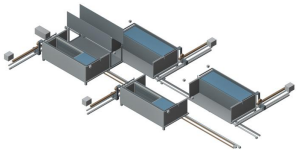
01 Guided lead screw



Selected architecture lineage

500 mm

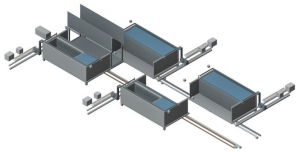
02 Rack and pawl



Runner-up

500 mm

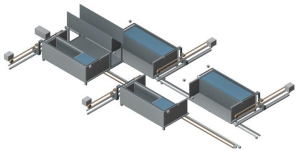
03 Spring and two gates



Retained alternative

500 mm

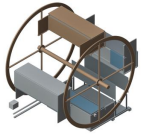
04 Belt shuttle



Retained alternative

500 mm

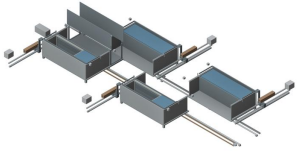
05 Indexed drum



Retained common-port concept

500 mm

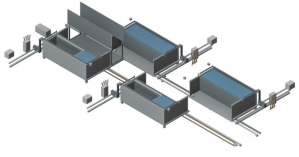
06 Telescopic carriage



Retained compact-stowage concept

500 mm

07 Pantograph



Retained linkage concept

500 mm

Shared orthographic view
Shared physical scale
Individually centered
Blue: payload envelope
Copper: drive geometry
Grey: structure/custody
Pale: actuator envelopes

Geometry: historical FDR09-R1. Trade roles: later FDR10-R0 / D27-R2 selection record.

Guided lead screw was developed further in FM08-R3. Current twelve-3U ACR1 is a separate assembly.